

Elimination of the Chemical Equilibrium Reaction Rates

The approach used is similar to those described in ([Lichtner, 1985](#), [Lichtner, 1988](#), and [Sevougian et al, 1992](#)). Consider the chemical-equilibrium reactions in the aqueous phase Equation ([D6.1](#)). Here $n_{aq} = n_c + n_a$ is the total number of components in the aqueous phase (gaseous solutes and aqueous components). It can be shown that for R_{aq} independent chemical reactions involving n_{aq} components ($n_{aq} > R_{aq}$), a canonical stoichiometric matrix $\mathbf{V}_c$ of dimension $n_{aq} \times R_{aq}$ that has the following form can be defined:

$$\mathbf{V}_c = \begin{bmatrix} \nu_{11} & \nu_{12} & \cdots & \nu_{1R_{aq}} \\ \vdots & \vdots & & \vdots \\ \nu_{n_{cp}1} & \nu_{n_{cp}2} & \cdots & \nu_{n_{cp}R_{aq}} \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (\text{D11.1})$$

Column j ($j = 1, \dots, R_{aq}$) of $\mathbf{V}_c$ corresponds to the stoichiometry of reaction j , i.e.,

$$\sum_{i=1}^{n_c} \nu_{ij} A_i + A_{n_c+j} = 0 \quad (\text{D11.2})$$

Equations (D11.1) and (D11.2) show that each of the n_{cp} components corresponding the first n_{cp} rows of $\mathbf{V}_c$ can participate in more than one of the independent chemical reactions Equation ([D6.1](#)). Each of the components corresponding to the last $R_{aq} = n_{aq} - n_{cp}$ rows of $\mathbf{V}_c$ participates only in one independent chemical reaction. The first n_{cp} components are called primary components and the last R_{aq} components are called secondary components. The gaseous components are always primary components.

The canonical stoichiometric matrix in Equation ([D11.1](#)) can be partitioned as follows:

$$\mathbf{V}_c = \begin{bmatrix} (\widetilde{\mathbf{V}})_{n_{cp} \times R_{aq}} \\ \mathbf{I}_{R_{aq}} \end{bmatrix} \quad (\text{D11.3})$$

where $\mathbf{I}_{R_{aq}}$ is the identity matrix of dimensions $R_{aq} \times R_{aq}$. For the R_{aq} chemical-equilibrium reactions, the forward rates are equal to the backward rates. There exists an *Equilibrium-Rate-Annihilation* (ERA) matrix ([Sevougian et al, 1992](#)) $\mathbf{E}$ of dimension $n_c \times n_{aq}$ such that the left $n_c \times n_c$ submatrix of $\mathbf{E}$ is non-singular and:

$$\mathbf{E} \mathbf{V}_c = \mathbf{0}_{n_{cp} \times R_{aq}} \quad (\text{D11.4})$$

It can be seen that

$$\mathbf{E} = \begin{bmatrix} \mathbf{I}_{n_{cp}} & (-\widetilde{\mathbf{V}})_{n_c \times R_{aq}} \end{bmatrix} \quad (\text{D11.5})$$

satisfies Equation [\(D11.4\)](#), and $\mathbf{I}_{n_{cp}}$ is non-singular. In simple terms, there are independent linear combinations of the component concentrations that have zero net reaction rates due to the chemical equilibrium reactions. This property can be used to reduce the number of component balance equations.

Let $\vec{\psi}$ be the vector of material balance equations for the gaseous and aqueous species Equations [\(D10.1\)](#) and [\(D10.2\)](#). Using Equation [\(D11.4\)](#), it can be shown that the product $\mathbf{E} \vec{\psi}$ gives combinations of component material balance equations that do not have terms with explicit chemical-equilibrium reactions rates $\sigma_{j,aq}$. The application of the ERA matrix reduces the number of flow equations from n_{aq} to $n_{aq} - R_{aq}$.

The elimination of the chemical-equilibrium reactions is important as it reduces the number of flow equations. If these chemical-equilibrium reactions are not eliminated but actually modeled with fast chemical reaction rates, the equations are more difficult to solve ([Litchner, 1991](#), [Litchner, 1992](#), and [Litchner, 1993](#)). The application of the ERA matrix reduces the stiffness of the equations and thus makes the solution easier to obtain.

Similarly, keyword ***MINERAL-ERA *ON** (default) creates an ERA matrix that includes the mineral reactions as well. Consider a stoichiometric matrix that includes both the chemical equilibrium and rate reactions. Index i of the stoichiometric matrix is from 1 to $(R_{aq} +$

$R_{mnr}) = R_{reac}$ where R_{mnr} is the number of mineral reactions. Index j of the stoichiometric matrix is from 1 to $(n_{aq} + n_{mnr}) = n_t$. The procedure outlined above is carried out in order to create a new ERA. However, the size of the ERA matrix is $(n_t - R_{reac}) \times n_t$. This new ERA matrix is multiplied with the material balance error so that the reaction terms cancel each other.